

## MTH 204 : Lecture 22

From Last time: Let  $f(x)$  be a periodic function of period  $2\pi$  and it can be represented by  $a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$  (i.e. the series converges and has sum  $f(x)$ )

then the series is called the Fourier Series Representation of  $f(x)$  and

Euler's Formula  $\left\{ \begin{array}{l} a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) dx, \quad a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(nx) dx \text{ for } n=1,2,\dots \\ \text{and } b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin(nx) dx \text{ for } n=1,2,\dots \end{array} \right.$

(Note: If the above series doesn't converge or doesn't converge to  $f(x)$ , it will be called the Fourier Series corresponding to  $f(x)$ .)

- We also checked the orthogonality relation between the functions  $1, \cos x, \sin x, \cos 2x, \sin 2x, \dots, \cos nx, \sin nx, \dots$

### Proof of Euler Formulas:

Suppose the function  $f(x)$  has a Fourier

Series representation as:

$$f(x) = a_0 + \sum_{n=1}^{\infty} [a_n \cos(nx) + b_n \sin(nx)]$$

$$\text{Then } \int_{-\pi}^{\pi} f(x) dx = \int_{-\pi}^{\pi} \left[ a_0 + \sum_{n=1}^{\infty} (a_n \cos(nx) + b_n \sin(nx)) \right] dx$$

$$= a_0 \int_{-\pi}^{\pi} dx + \sum_{n=1}^{\infty} \left( a_n \int_{-\pi}^{\pi} \cos(nx) dx + b_n \int_{-\pi}^{\pi} \sin(nx) dx \right)$$

$$= a_0 \times 2\pi + 0 = 2\pi a_0$$

$$\text{So, } \boxed{a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) dx} = \frac{\langle f(x), 1 \rangle}{\langle 1, 1 \rangle}$$

$$\text{Now } \int_{-\pi}^{\pi} f(x) \cos(mx) dx = \int_{-\pi}^{\pi} \left[ a_0 + \sum_{n=1}^{\infty} (a_n \cos(nx) + b_n \sin(nx)) \right] \cos(mx) dx$$

$$= a_0 \int_{-\pi}^{\pi} \cos(mx) dx + \sum_{n=1}^{\infty} \left[ a_n \int_{-\pi}^{\pi} \cos(nx) \cos(mx) dx + b_n \int_{-\pi}^{\pi} \sin(nx) \cos(mx) dx \right]$$

$$= 0 + a_m \times \pi + 0 = a_m \pi$$

$$\Rightarrow \boxed{a_m = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(mx) dx} = \frac{\langle f(x), \cos(mx) \rangle}{\langle \cos(mx), \cos(mx) \rangle}$$

$$\int_{-\pi}^{\pi} f(x) \sin(mx) dx = \int_{-\pi}^{\pi} \left[ a_0 + \sum_{n=1}^{\infty} (a_n \cos(nx) + b_n \sin(nx)) \right] \sin(mx) dx$$

$$= a_0 \int_{-\pi}^{\pi} \sin(mx) dx + \sum_{n=1}^{\infty} \left[ a_n \int_{-\pi}^{\pi} \cos(nx) \sin(mx) dx + b_n \int_{-\pi}^{\pi} \sin(nx) \sin(mx) dx \right]$$

$$= 0 + 0 + b_m \times \pi = b_m \pi$$

$$\Rightarrow \boxed{b_m = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin(mx) dx} = \frac{\langle f(x), \sin(mx) \rangle}{\langle \sin(mx), \sin(mx) \rangle}$$

## Representation by a Fourier Series:

Let  $f(x)$  be periodic with period  $2\pi$  and piecewise continuous in the interval  $[-\pi, \pi]$ .

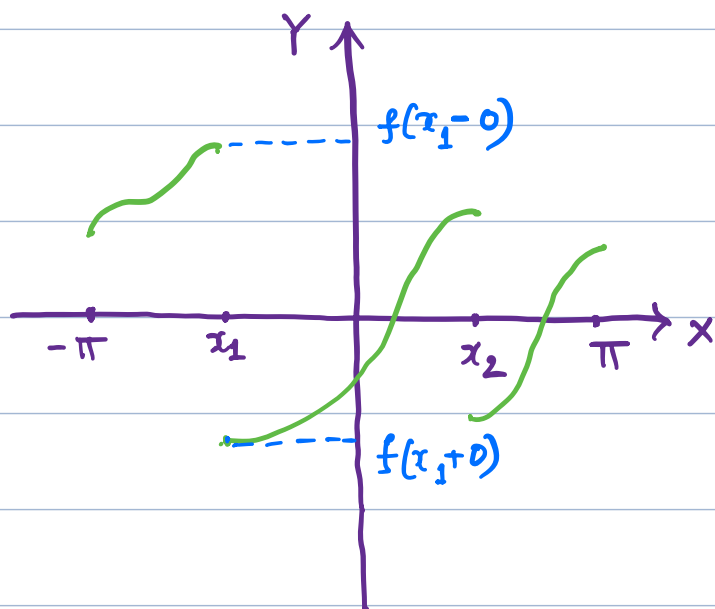
Furthermore, let  $f(x)$  have a left-hand derivative and a right-hand derivative at each point of that interval. Then the Fourier series of  $f(x)$  converges. Its sum is  $f(x)$ , except at points  $x_0$  where  $f(x)$  is discontinuous. There the sum of the series is the average of the left- and right-hand limits of  $f(x)$  at  $x_0$  i.e.  $\frac{f(x_0+0) + f(x_0-0)}{2}$

$$\text{Here } f(x_0+0) = \lim_{\substack{h \rightarrow 0 \\ h > 0}} f(x_0+h)$$

$$f(x_0 - h) = \lim_{\substack{h \rightarrow 0 \\ h > 0}} f(x_0 - h)$$

$$f'(x_0 + 0) = \lim_{\substack{h \rightarrow 0 \\ h > 0}} \frac{f(x_0 + h) - f(x_0 + 0)}{h}$$

$$f'(x_0 - 0) = \lim_{\substack{h \rightarrow 0 \\ h > 0}} \frac{f(x_0 - h) - f(x_0 - 0)}{-h}$$



• Sum of the fourier series is  $f(x)$  where  $f$  is continuous

• At  $x_1$ , the sum is  $\frac{f(x_1+0) + f(x_1-0)}{2}$

• At  $x_2$ , the sum is  $\frac{f(x_2+0) + f(x_2-0)}{2}$

## Orthogonal Decomposition:

Let  $W$  be a vector subspace of a vector space  $V$ . Then any vector  $y \in V$  can be

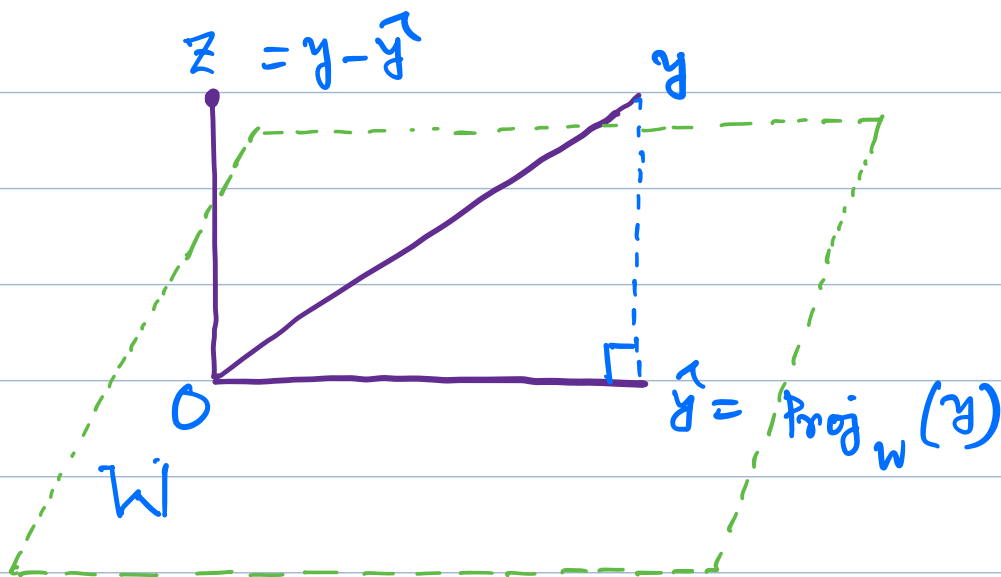


written uniquely as  $y = \hat{y} + z$   
 where  $\hat{y} \in W$  and  $z \in W^\perp$ .

If  $\{u_1, u_2, \dots, u_p\}$  (or  $\{u_1, u_2, \dots\}$  if  $W$  is infinite dimensional)  
 is an orthogonal basis of  $W$ ,

then 
$$\hat{y} = \frac{\langle y, u_1 \rangle}{\|u_1\|^2} u_1 + \frac{\langle y, u_2 \rangle}{\|u_2\|^2} u_2$$

$$+ \dots + \frac{\langle y, u_p \rangle}{\|u_p\|^2} u_p \quad \left( \begin{array}{l} \text{a series} \\ \text{in case } W \\ \text{is infinite} \\ \text{dimensional} \end{array} \right)$$



Fourier Series can be viewed as an Orthogonal Projection.

$$f(x) = a_0 + \sum_{n=1}^{\infty} [a_n \cos(nx) + b_n \sin(nx)]$$

$$\text{where } a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) dx = \frac{\langle f(x), 1 \rangle}{\|1\|^2}$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(nx) dx = \frac{\langle f(x), \cos(nx) \rangle}{\|\cos(nx)\|^2}$$

$$\text{and } b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin(nx) dx = \frac{\langle f(x), \sin(nx) \rangle}{\|\sin(nx)\|^2}$$

## Periodic Functions with arbitrary period:

Suppose  $f(x)$  is periodic function with period  $p = 2L$  (i.e.  $f(x+2L) = f(x)$ )

Then we can introduce a new variable  $v$

$$\text{Such that } v = \frac{2\pi}{p} x = \frac{2\pi}{2L} x = \frac{\pi}{L} x$$

$$\Rightarrow x = \frac{L}{\pi} v$$

Then the function  $f(x) = f\left(\frac{L}{\pi} v\right)$

as a function of  $v$  is a periodic function with period  $2\pi$

( $v = \pm\pi$  corresponds to  $x = \pm L$ )

Thus  $f(x) = f\left(\frac{L}{\pi} v\right)$  has a fourier series of the form:

$$f(x) = f\left(\frac{L}{\pi} v\right) = a_0 + \sum_{n=1}^{\infty} (a_n \cos nv + b_n \sin nv)$$

$$= a_0 + \sum_{n=1}^{\infty} \left[ a_n \cos\left(\frac{n\pi}{L} x\right) + b_n \sin\left(\frac{n\pi}{L} x\right) \right]$$

$$\text{where } a_0 = \frac{1}{2L} \int_{-L}^L f(x) dx$$

$$a_n = \frac{1}{L} \int_{-L}^L f(x) \cos\left(\frac{n\pi}{L} x\right) dx \quad \text{for } n=1,2,\dots$$

$$b_n = \frac{1}{L} \int_{-L}^L f(x) \sin\left(\frac{n\pi}{L} x\right) dx \quad \text{for } n=1,2,\dots$$

Check: If  $f(x) = f\left(\frac{L}{\pi} v\right) = a_0 + \sum_{n=1}^{\infty} (a_n \cos(nv) + b_n \sin(nv))$

then  $a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} f\left(\frac{L}{\pi} v\right) dv = \frac{1}{2\pi} \int_{-L}^L f(x) \frac{\pi}{L} dx$

$\left( \begin{array}{l} v = \frac{\pi}{L} x \Rightarrow dv = \frac{\pi}{L} dx \\ v = \pm \pi \Rightarrow x = \pm L \end{array} \right) = \frac{1}{2L} \int_{-L}^L f(x) dx$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f\left(\frac{L}{\pi} v\right) \cos(nv) dv$$

$$= \frac{1}{\pi} \int_{-L}^L f(x) \cos\left(\frac{n\pi}{L} x\right) \frac{\pi}{L} dx$$

$$= \frac{1}{L} \int_{-L}^L f(x) \cos\left(\frac{n\pi}{L} x\right) dx$$

and  $b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f\left(\frac{L}{\pi} v\right) \sin(nv) dv$

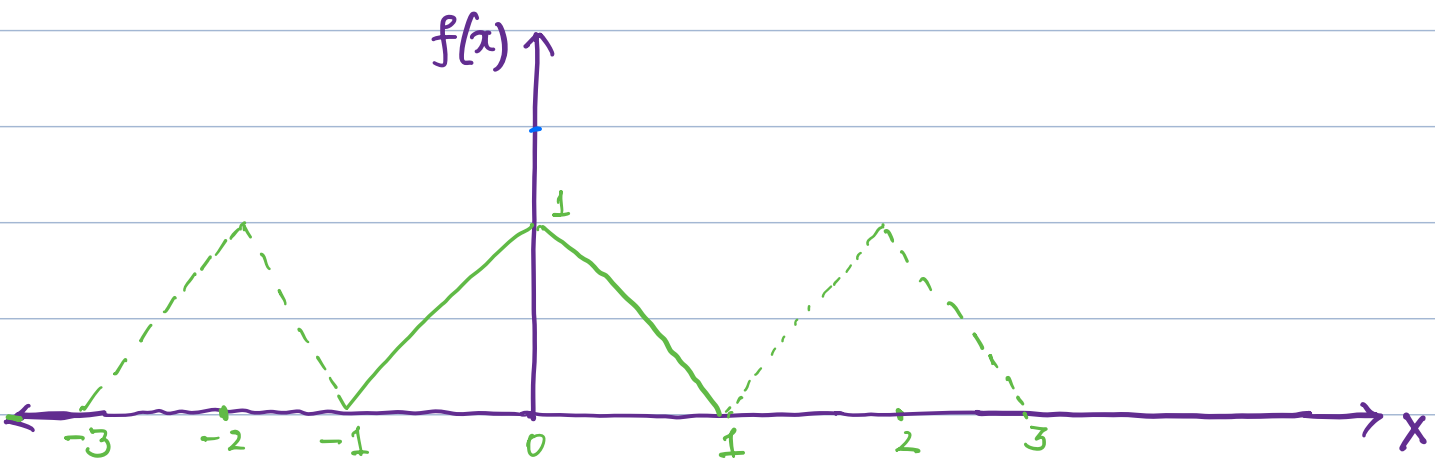
$$= \frac{1}{\pi} \int_{-L}^L f(x) \sin\left(\frac{n\pi}{L} x\right) \frac{\pi}{L} dx$$

$$= \frac{1}{L} \int_{-L}^L f(x) \sin\left(\frac{n\pi}{L} x\right) dx$$

- The set  $\left\{ 1, \cos\left(\frac{\pi}{L} x\right), \sin\left(\frac{\pi}{L} x\right), \cos\left(\frac{2\pi}{L} x\right), \sin\left(\frac{2\pi}{L} x\right), \dots, \cos\left(\frac{n\pi}{L} x\right), \sin\left(\frac{n\pi}{L} x\right), \dots \right\}$  forms an orthogonal basis of the vector space of  $2L$ -periodic functions.

Ex:

$$f(x) = \begin{cases} 1+x & \text{if } -1 < x < 0 \\ 1-x & \text{if } 0 < x < 1 \end{cases} \quad \left. \vphantom{f(x)} \right\} \text{Period } p = 2$$



$$\text{Period } p = 2L = 2 \Rightarrow L = 1$$

$$\text{Now } a_0 = \frac{1}{2L} \int_{-L}^L f(x) dx = \frac{1}{2} \int_{-1}^1 f(x) dx$$

$$= \frac{1}{2} \left[ \int_{-1}^0 (1+x) dx + \int_0^1 (1-x) dx \right] = \frac{1}{2} \left\{ \left[ x + \frac{x^2}{2} \right]_{-1}^0 + \left[ x - \frac{x^2}{2} \right]_0^1 \right\}$$

$$= \frac{1}{2} \left\{ \left(1 - \frac{1}{2}\right) + \left(1 - \frac{1}{2}\right) \right\} = \boxed{\frac{1}{2}}$$

$$\begin{aligned} a_n &= \frac{1}{1} \int_{-1}^1 f(x) \cos\left(\frac{n\pi}{L}x\right) dx = \int_{-1}^0 (1-x) \cos(n\pi)x dx + \int_0^1 (1+x) \cos(n\pi)x dx \\ &= \left[ \frac{\sin(n\pi)x}{n\pi} + \frac{x}{n\pi} \sin(n\pi)x + \frac{\cos(n\pi)x}{n^2\pi^2} \right]_{-1}^0 \\ &\quad + \left[ \frac{\sin(n\pi)x}{n\pi} - \frac{x \sin(n\pi)x}{n\pi} - \frac{\cos(n\pi)x}{n^2\pi^2} \right]_0^1 \\ &= \frac{1}{n^2\pi^2} - \frac{(-1)^n}{n^2\pi^2} - \frac{(-1)^n}{n^2\pi^2} + \frac{1}{n^2\pi^2} = \frac{2[1 - (-1)^n]}{n^2\pi^2} \end{aligned}$$

$$\begin{aligned} b_n &= \frac{1}{L} \int_{-L}^L f(x) \sin\left(\frac{n\pi}{L}x\right) dx \\ &= \int_{-1}^0 (1-x) \sin(n\pi)x dx + \int_0^1 (1+x) \sin(n\pi)x dx \\ &= \left[ -\frac{\cos(n\pi)x}{n\pi} + \frac{x \cos(n\pi)x}{n\pi} - \frac{\sin(n\pi)x}{n^2\pi^2} \right]_{-1}^0 \\ &\quad + \left[ -\frac{\cos(n\pi)x}{n\pi} - \frac{x \cos(n\pi)x}{n\pi} + \frac{\sin(n\pi)x}{n^2\pi^2} \right]_0^1 \end{aligned}$$

$$= -\cancel{\frac{1}{n\pi}} + \cancel{\frac{\cos(n\pi)}{n\pi}} + \cancel{\frac{\cos(n\pi)}{n\pi}} - \cancel{\frac{\cos(n\pi)}{n\pi}} - \cancel{\frac{\cos(n\pi)}{n\pi}} + \cancel{\frac{1}{n\pi}} = 0$$

$$\text{So, } f(x) = \frac{1}{2} + \sum_{n=1}^{\infty} \frac{2[1 - (-1)^n]}{n^2\pi^2} \cos(n\pi)x$$

$$= \boxed{\frac{1}{2} + \frac{4}{\pi^2} \left[ \frac{\cos x}{1} + \frac{\cos(3\pi x)}{9} + \frac{\cos(5\pi x)}{25} + \dots \right]}$$

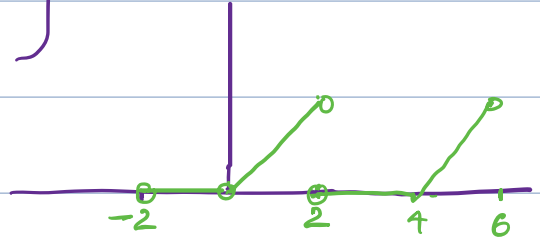
Since the function is even, it has a cosine only series.

Ex: Find the Fourier Series of

$$\left. \begin{aligned} f(x) &= 0 & \text{for } -2 < x < 0 \\ x & & \text{for } 0 < x < 2 \end{aligned} \right\} \text{ with period } p=4$$

$$\text{So, } p = 2L = 4 \Rightarrow L = 2$$

$$a_0 = \frac{1}{2L} \int_{-L}^L f(x) dx = \frac{1}{4} \int_{-2}^2 f(x) dx = \frac{1}{4} \int_0^2 x dx = \frac{1}{4} \left[ \frac{x^2}{2} \right]_0^2 = \frac{1}{4} \cdot \frac{4}{2} = \boxed{\frac{1}{2}}$$



$$a_n = \frac{1}{2} \int_{-L}^L f(x) \cos\left(\frac{n\pi x}{L}\right) dx = \frac{1}{2} \int_0^2 x \cos\left(\frac{n\pi}{2} x\right) dx$$

$$= \frac{1}{2} \left[ \frac{x \sin\left(\frac{n\pi}{2} x\right)}{\frac{n\pi}{2}} + \frac{\cos\left(\frac{n\pi}{2} x\right)}{\frac{n^2 \pi^2}{4}} \right]_0^2 = \boxed{\frac{2}{n^2 \pi^2} [(-1)^n - 1]}$$

$$\Rightarrow \left. \begin{aligned} a_n &= 0 & \text{if } n \text{ is even.} \\ &= -\frac{4}{n^2 \pi^2} & \text{if } n \text{ is odd.} \end{aligned} \right\}$$

$$b_n = \frac{1}{2} \int_0^2 x \sin\left(\frac{n\pi}{2} x\right) dx = \frac{1}{2} \left[ -x \frac{\cos\left(\frac{n\pi}{2} x\right)}{\left(\frac{n\pi}{2}\right)} + \frac{\sin\left(\frac{n\pi}{2} x\right)}{\frac{n^2 \pi^2}{4}} \right]_0^2$$

$$= \frac{-2 \cos(n\pi)}{n\pi} = \frac{-2 (-1)^n}{n\pi} = \boxed{\frac{2 (-1)^{n+1}}{n\pi}}$$

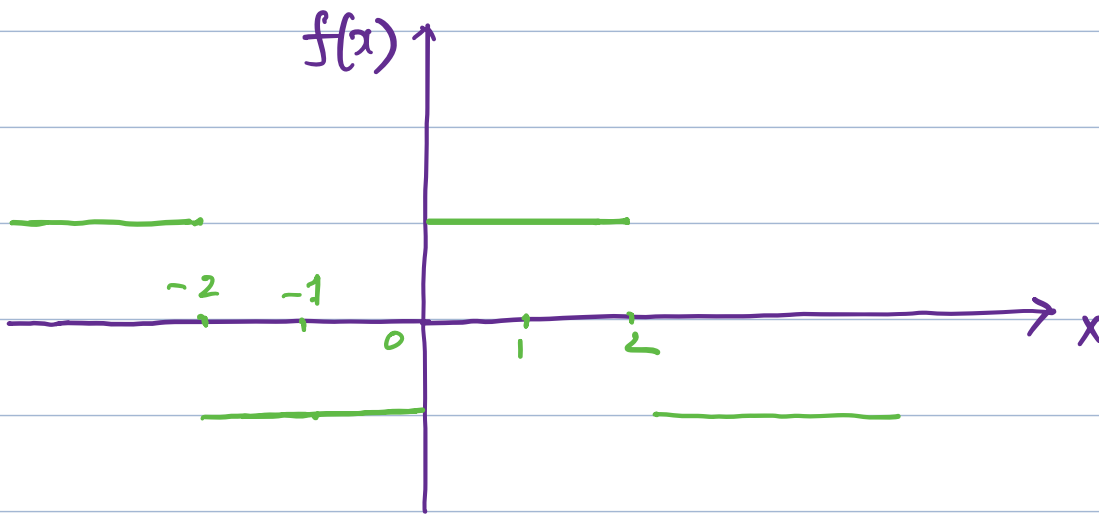
So, the Fourier Series is:

$$\begin{aligned} f(x) &= \frac{1}{2} - \frac{4}{\pi^2} \left[ \cos\left(\frac{\pi}{2} x\right) + \frac{1}{9} \cos\left(\frac{3\pi}{2} x\right) + \frac{1}{25} \cos\left(\frac{5\pi}{2} x\right) \right. \\ &\quad \left. + \dots \right] \\ &\quad + \frac{2}{\pi} \left[ \sin\left(\frac{\pi}{2} x\right) - \frac{1}{2} \sin\left(\frac{2\pi}{2} x\right) + \frac{1}{3} \sin\left(\frac{3\pi}{2} x\right) \right. \\ &\quad \left. - \dots \right] \end{aligned}$$

Ex: Change of Scale:

$$\text{Let } \begin{cases} f(x) = -k & -2 < x < 0 \\ = k & 0 < x < 2 \end{cases}$$

$$\text{and } f(x+4) = f(x)$$



$$\text{The period } p = 2L = 4 \Rightarrow L = 2$$

Now from a previous example: If  $g(x)$  is a periodic function with period  $2\pi$

$$g(v) = \begin{cases} -k & \text{if } -\pi < v < 0 \\ k & \text{if } 0 < v < \pi \end{cases}$$

$$= \sum_{n=1}^{\infty} \frac{2k}{n\pi} [1 - (-1)^n] \sin nv$$



$$= \frac{4k}{\pi} \sin v + \frac{4\pi}{3\pi} \sin(3v) + \frac{4\pi}{5\pi} \sin(5v) + \dots$$

Now let  $v = \frac{\pi}{2} x$

So,  $x = \frac{2v}{\pi}$  and  $v = \pm\pi \Rightarrow x = \pm 2$

Thus  $f(x) = g(v) = g\left(\frac{\pi}{2} x\right)$

$$= \frac{4k}{\pi} \sin\left(\frac{\pi}{2} x\right) + \frac{4k}{3\pi} \sin\left(\frac{3\pi}{2} x\right) + \frac{4k}{5\pi} \sin\left(\frac{5\pi}{2} x\right) + \dots$$

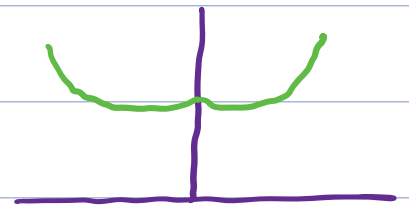
### Even and Odd functions:

- If  $f(x)$  is an even function

i.e.  $f(-x) = f(x)$

its Fourier series reduces to a Fourier cosine series

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi}{L} x\right)$$

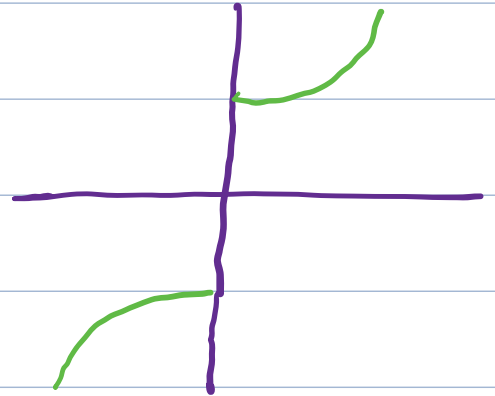


where  $a_0 = \frac{1}{L} \int_0^L f(x) dx$

and  $a_n = \frac{2}{L} \int_0^L f(x) \cos\left(\frac{n\pi x}{L}\right) dx, \quad n=1, 2, \dots$

- If  $f(x)$  is an odd function that is  $f(-x) = -f(x)$ , then its Fourier series reduces to a Fourier sine series

$$f(x) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi}{L} x\right)$$



where  $b_n = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi}{L} x\right) dx$

We have used  $\int_{-L}^L g(x) dx = 2 \int_0^L g(x) dx$  if  $g$  is even

and  $\int_{-L}^L h(x) dx = 0$  if  $h$  is odd

Theorem: (Sum and scalar multiple)

The Fourier coefficients of a sum  $f_1 + f_2$  are the sums of the corresponding Fourier coefficients of  $f_1$  and  $f_2$

The Fourier coefficients of  $cf$  are  $c$  times the corresponding Fourier coefficients of  $f$ .

In other words we can say that:

The process of obtaining Fourier series is linear.

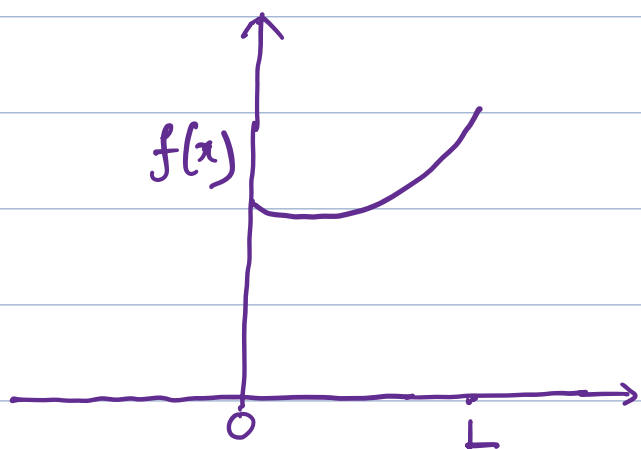
Let  $FS$  be an operator that assigns to each function, its Fourier series.

$$\text{Then } FS(f_1 + f_2) = FS(f_1) + FS(f_2)$$

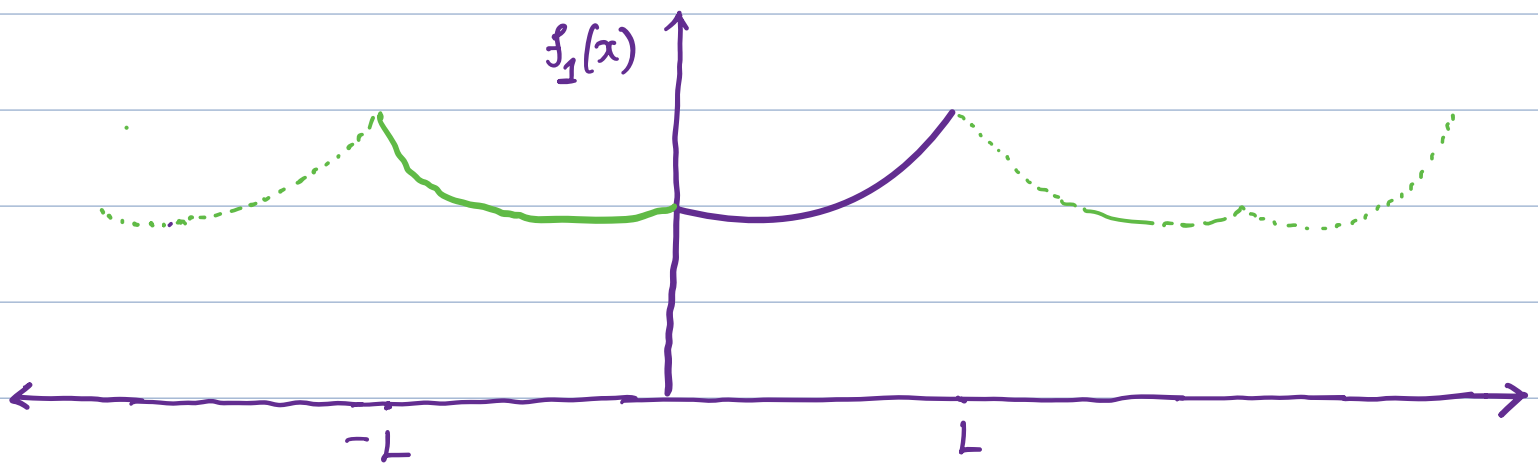
$$FS(cf) = cFS(f)$$

### Half-range Expansion

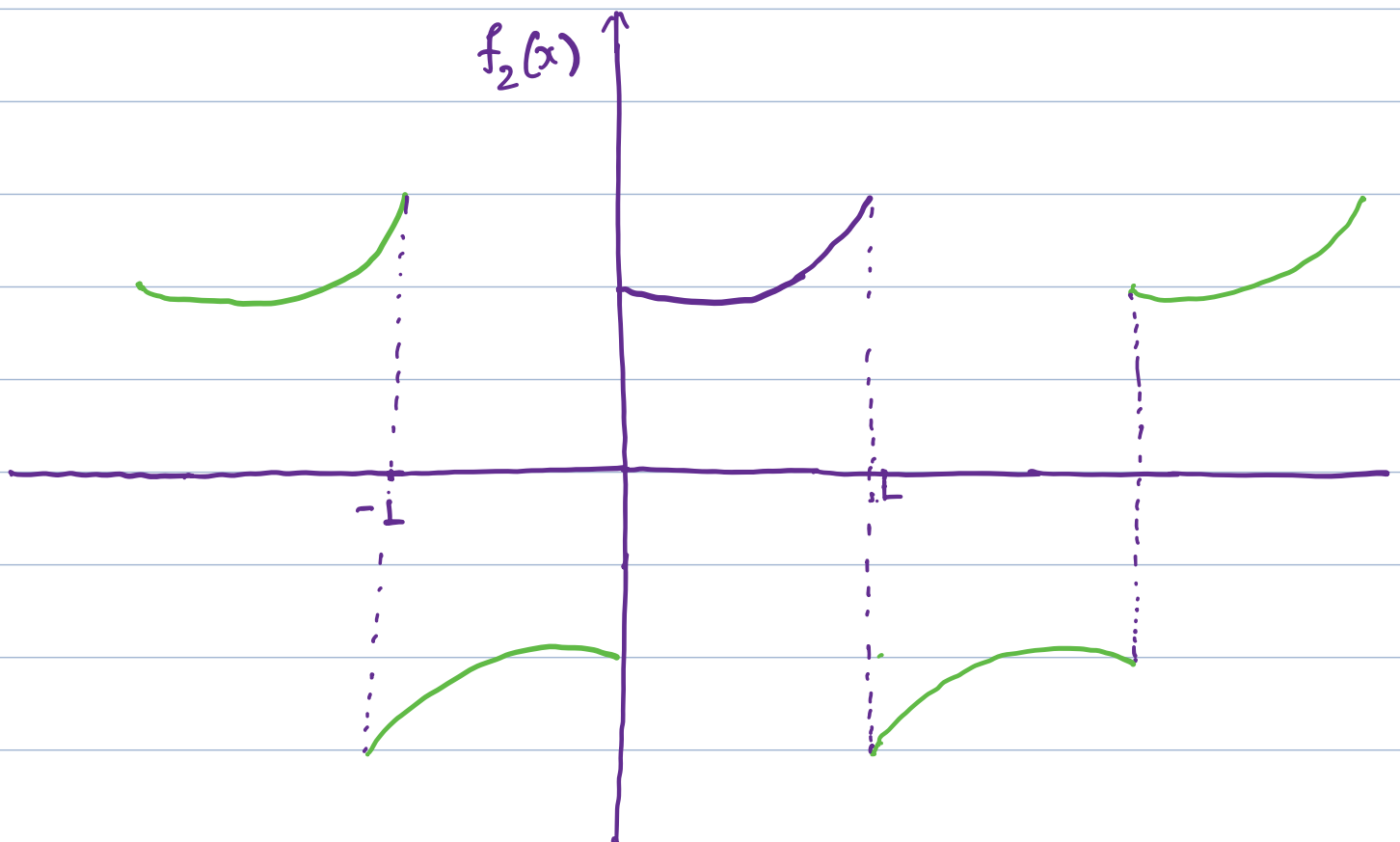
If only half of the range  $[0, L]$  is of interest, we may extend, we may extend the function in an odd or even way and then use the simplified Fourier series expression for odd and even function.



(The given function  $f(x)$ )

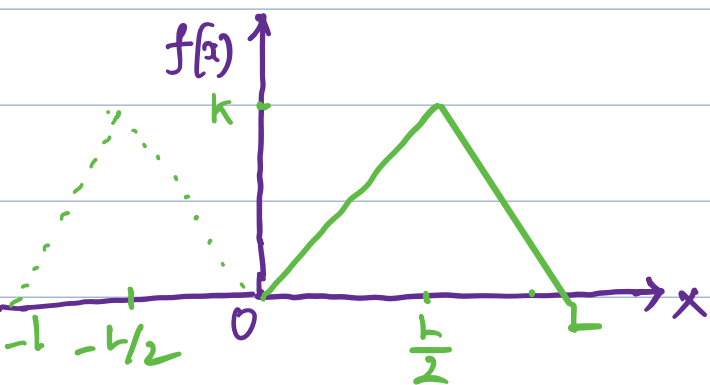


$\left( f(x) \text{ continued as an even periodic function of period } 2L \right)$



$\left( f(x) \text{ continued as an odd periodic function of period } 2L \right)$

Ex:  $f(x) = \frac{2k}{L} x$  for  $0 < x < \frac{L}{2}$   
 $= \frac{2k}{L} (L-x)$  for  $\frac{L}{2} < x < L$



Even extension:  $a_0 = \frac{1}{L} \int_0^L f(x) dx = \frac{1}{L} \cdot \frac{1}{2} L k = \frac{k}{2}$

$$a_n = \frac{2}{L} \int_0^L f(x) \cos\left(\frac{n\pi}{L} x\right) dx$$

$$= \frac{2}{L} \left[ \frac{2k}{L} \int_0^{L/2} x \cos\left(\frac{n\pi}{L} x\right) dx + \frac{2k}{L} \int_{L/2}^L (L-x) \cos\left(\frac{n\pi}{L} x\right) dx \right]$$

$$= \frac{4k}{n^2 \pi^2} \left[ 2 \cos\left(\frac{n\pi}{2}\right) - \cos(n\pi) - 1 \right] \quad \left( \text{using integration by parts} \right)$$

So,  $\tilde{f}_e(x) = \frac{k}{2} - \frac{16k}{\pi^2} \left( \frac{1}{2^2} \cos\left(\frac{2\pi}{L} x\right) + \frac{1}{6^2} \cos\left(\frac{6\pi}{L} x\right) + \dots \right)$

Odd extension:

$$b_n = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi}{L} x\right) dx$$

$$= \frac{2}{L} \left[ \frac{2k}{L} \int_0^{L/2} x \sin\left(\frac{n\pi}{L} x\right) dx + \frac{2k}{L} \int_{L/2}^L (L-x) \sin\left(\frac{n\pi}{L} x\right) dx \right]$$

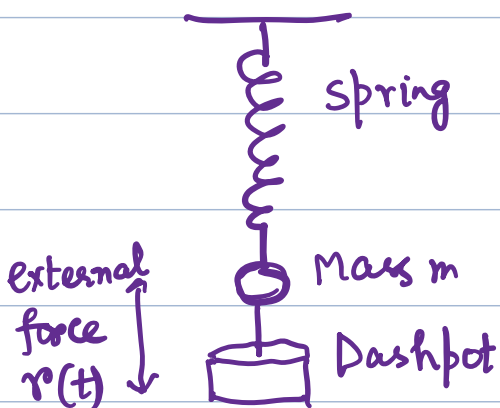
$$= \frac{8k}{n^2 \pi^2} \sin\left(\frac{n\pi}{2}\right) \quad \left(\text{Using integration by parts}\right)$$

Therefore

$$\tilde{f}_0(x) = \frac{8k}{\pi^2} \left( \frac{1}{1^2} \sin\left(\frac{\pi}{L}x\right) - \frac{1}{3^2} \sin\left(\frac{3\pi}{L}x\right) + \frac{1}{5^2} \sin\left(\frac{5\pi}{L}x\right) - \dots \right)$$

## Forced oscillations.

Example: undamped string:

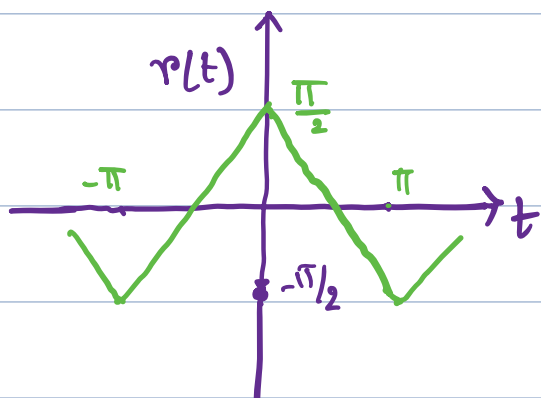


$$my'' + cy' + ky = r(t)$$

Given:  $m = 1(\text{gm})$

$$c = .05(\text{gm/sec})$$

$$k = 25(\text{gm/sec}^2)$$



Then  $y'' + .05y' + 25y = r(t)$

Given:  $r(t) = t + \frac{\pi}{2}$  for  $-\pi < t < 0$   
 $= -t + \frac{\pi}{2}$  for  $0 < t < \pi$

We expand the driving force in its fourier series

$$r(t) = \frac{4}{\pi} \left( \cos t + \frac{1}{3^2} \cos(3t) + \frac{1}{5^2} \cos(5t) + \dots \right)$$

Consider the ODE:  $y'' + .05y' + 25y = \frac{4}{\pi n^2} \cos(nt)$

The steady state solution is of the form  $y_n = A_n \cos(nt) + B_n \sin(nt)$

Substituting in the equation we get  $A_n = \frac{4(25-n^2)}{n^2 \pi D_n}$

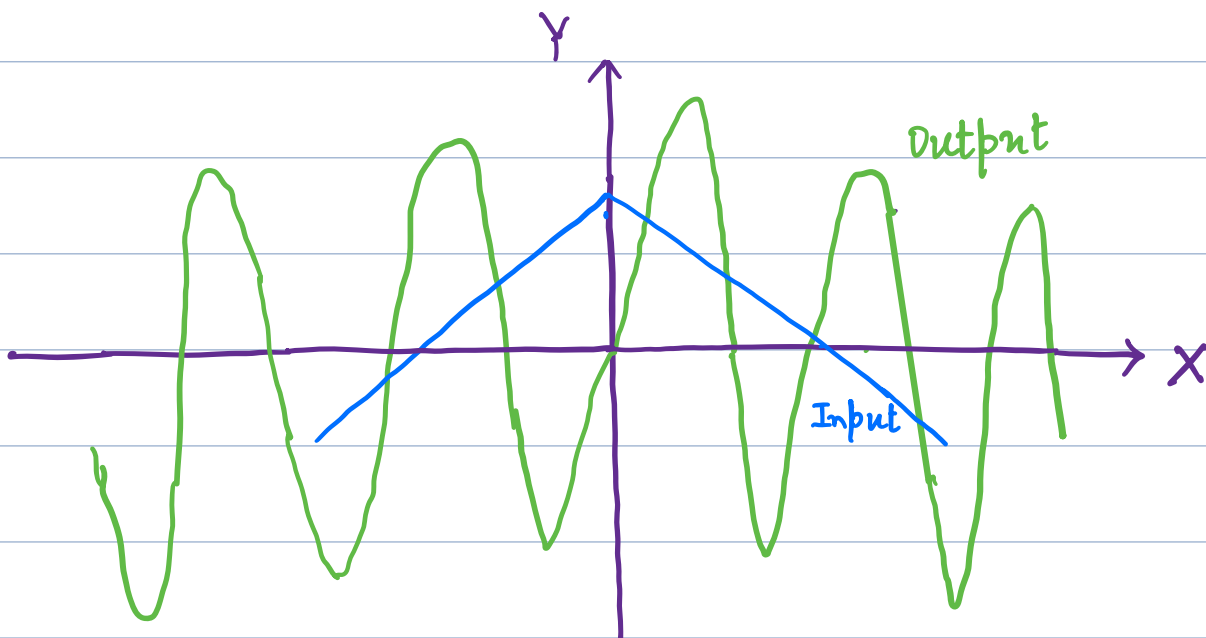
and  $B_n = \frac{0.2}{n \pi D_n}$  where  $D_n = (25-n^2)^2 + (.05n)^2$

The solution is:

$$y_n = \frac{4(25-n^2)}{n^2 \pi D_n} \cos(nt) + \frac{0.2}{n \pi D_n} \sin(nt)$$

The steady state solution of the problem for the given  $r(t)$  is:

$$y = y_1 + y_3 + y_5 + \dots$$



# Approximation by Trigonometric Polynomials

Let  $f(x)$  be a function on the interval  $[-\pi, \pi]$  that can be represented on this interval by a Fourier series.

Then the  $N$ th partial sum of the Fourier series

$$a_0 + \sum_{n=1}^N [a_n \cos(nx) + b_n \sin(nx)]$$

is an approximation of the given  $f(x)$ .

If we choose an arbitrary  $N$  and keep it fixed, then we can ask for the "best" approximation of  $f$  by a Trigonometric Polynomial of the same degree  $N$  that is by a function of the form:

$$F(x) = A_0 + \sum_{n=1}^N [A_n \cos(nx) + B_n \sin(nx)]$$

( $N$  is fixed)

We choose the error of approximation as:

$$E = \int_{-\pi}^{\pi} (f - F)^2 dx$$

This is called the square error of  $F$  relative to the function  $f$  on  $[-\pi, \pi]$ .



Clearly  $E \geq 0$

$$\text{Now, } E = \int_{-\pi}^{\pi} f^2 dx - 2 \int_{-\pi}^{\pi} f F dx + \int_{-\pi}^{\pi} F^2 dx$$

$$\text{Now } \int_{-\pi}^{\pi} F^2 dx = \int_{-\pi}^{\pi} \left[ A_0 + \sum_{n=1}^N [A_n \cos(nx) + B_n \sin(nx)] \right]^2 dx$$

$$= 2\pi A_0^2 + A_1^2 \pi + A_2^2 \pi + \dots + A_N^2 \pi \\ + B_1^2 \pi + B_2^2 \pi + \dots + B_N^2 \pi + 0$$

(By Orthogonality relation)

$$= \pi (2A_0^2 + A_1^2 + A_2^2 + \dots + A_N^2 + B_1^2 + B_2^2 + \dots + B_N^2)$$

$$\text{Now } \int_{-\pi}^{\pi} f F dx = \int_{-\pi}^{\pi} f \left[ A_0 + \sum_{n=1}^N [A_n \cos(nx) + B_n \sin(nx)] \right] dx$$

$$= A_0 \int_{-\pi}^{\pi} f dx + \sum_{n=1}^N \left[ A_n \int_{-\pi}^{\pi} f \cos(nx) dx + B_n \int_{-\pi}^{\pi} f \sin(nx) dx \right]$$

$$= \pi (2A_0 a_0 + A_1 a_1 + \dots + A_N a_N + B_1 b_1 + \dots + B_N b_N)$$

$$\text{Therefore } E = \int_{-\pi}^{\pi} f^2 dx - 2\pi \left[ 2A_0 a_0 + \sum_{n=1}^N (A_n a_n + B_n b_n) \right] \\ + \pi \left[ 2A_0^2 + \sum_{n=1}^N (A_n^2 + B_n^2) \right]$$

Now we choose  $A_n = a_n$  and  $B_n = b_n$  for  $n=0, 1, \dots, N$   
(i.e. we choose the  $n$ th partial sum of the Fourier series as our approximating polynomial)

we call the error for this particular Trigonometric polynomial as  $E^*$ .

$$\text{Then } E^* = \int_{-\pi}^{\pi} f^2 dx - \pi \left[ 2a_0^2 + \sum_{n=1}^N (a_n^2 + b_n^2) \right]$$

$$\text{So, } E - E^* = \pi \left[ 2(A_0 - a_0)^2 + \sum_{n=1}^N [(A_n - a_n)^2 + (B_n - b_n)^2] \right] \geq 0$$

Thus  $E \geq E^*$

and  $E = E^*$  if and only if  $A_0 = a_0, \dots, B_N = b_N$

Therefore we have the following result:

Minimum Square Error:

The square error of  $F$  (with fixed  $N$ ) relative to  $f$  on  $[-\pi, \pi]$  is minimum if and only if the coefficients of  $F$  are the Fourier coefficients of  $f$ .

The minimum error is given by

$$E^* = \int_{-\pi}^{\pi} f^2 dx - \pi \left[ 2a_0^2 + \sum_{n=1}^N (a_n^2 + b_n^2) \right]$$

Note:

- Since  $E^* > 0$  we get

$$2a_0^2 + \sum_{n=1}^N (a_n^2 + b_n^2) \leq \frac{1}{\pi} \int_{-\pi}^{\pi} f^2 dx$$

Since the above holds for every  $N$ , we obtain the

Bessel Inequality:

$$2a_0^2 + \sum_{n=1}^{\infty} (a_n^2 + b_n^2) \leq \frac{1}{\pi} \int_{-\pi}^{\pi} f^2 dx$$

for the Fourier coefficients of any function  $f$  for which the integral on the right exists.

- It can also be shown that for such a function  $f$  (for which  $\int_{-\pi}^{\pi} f^2 dx$  exists), equality holds in the above inequality and it becomes Parseval's identity:

$$2a_0^2 + \sum_{n=1}^{\infty} (a_n^2 + b_n^2) = \frac{1}{\pi} \int_{-\pi}^{\pi} f^2 dx$$

